

Pairs Trading Framework

Statistical Arbitrage on the NIFTY100 Universe

Quantitative Research

May 5, 2026

Agenda

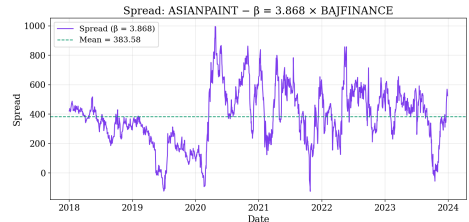
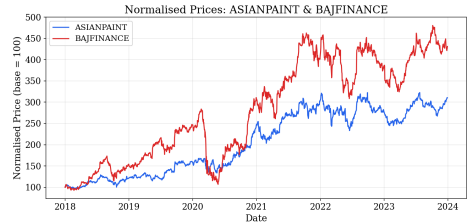
- 1 Motivation & Overview
- 2 Pair Selection (In-Sample)
- 3 Spread & Z-Score Models
- 4 Signal Generation
- 5 Prediction-based Strategies
- 6 Backtesting Framework
- 7 Baselines

What is Pairs Trading?

Core idea: Two historically co-moving assets temporarily diverge. We bet on reversion to their long-run equilibrium.

Three-step logic

- 1 Identify pairs
- 2 Model the stationary *spread* between them
- 3 Bet on the mean-reversion of the spread



- **Distance-based:** Euclidean distance between normalized (z-score) price series.
- **Correlation-based:** Pearson correlation of log returns.
- **Cointegration-based:** Engle-Granger or Johansen tests.

Cointegration-based methods are the most robust of the 3 for pairs trading, as they ensure mean-reversion. However, they are computationally intensive, especially for large universes like the NIFTY100.

Two-Stage Pair Filtering

- Cointegration tests (Engle-Granger, Johansen) are generally used to identify co-movement like that of ASIANPAINT and BAJFINANCE. However, they are computationally expensive.
- We use a two-stage approach:
 - ① **Correlation screening**
 - ② **Cointegration testing** on the screened pairs
- Then, the top k pairs with highest sharpe ratio on the training window are selected for trading.

Key insight

Correlation is a *necessary* but not *sufficient* condition. Cointegration guarantees a stationary spread — the actual trading signal.

Spread Construction & Hedge Ratio

For each selected pair (y, x) , the spread is:

$$\text{Spread}_t = y_t - \beta x_t$$

where β is the **hedge ratio**.

Static β

- OLS over full training window
- Simple, interpretable
- Assumes constant relationship
- Valid for small enough trading windows
- This is what we implemented

Dynamic β

- Rolling OLS or Kalman filter
- Adapts to structural shifts
- Higher complexity

Z-Score Parameterisations

Simple (Rolling 60-day)

$$Z_t = \frac{\text{Spread}_t - \mu_{60}}{\sigma_{60}}$$

Standard rolling mean and standard deviation over a 60-day window.

Ornstein-Uhlenbeck

Model: $dS_t = \theta(\mu - S_t) dt + \sigma dW_t$

Estimate via regression $\Delta S_t = a + b S_{t-1}$:

$$\hat{\theta} = -b, \quad \hat{\mu} = \frac{a}{1 - \hat{\theta}}$$

$$Z_t = \frac{\text{Spread}_t - \mathbb{E}_{OU}}{\text{Std}_{OU}}$$

The OU z-score accounts for the *expected* path of mean-reversion, making entry signals more statistically grounded than a simple rolling window.

Trading Signals

Condition	Position	Action
$Z_t > +2.0$	-1 (Short spread)	Short y, Long x
$Z_t < -2.0$	+1 (Long spread)	Long y, Short x
$ Z_t \leq 0.5$	0 (Flat)	Close position
Stop-loss hit	0 (Flat)	Emergency exit
Otherwise	Unchanged	Hold

Portfolio aggregation

Capital is **equally weighted** across all k active pairs. Portfolio wealth carries forward continuously across walk-forward windows.

Krauss et al.'s (2017) approach:

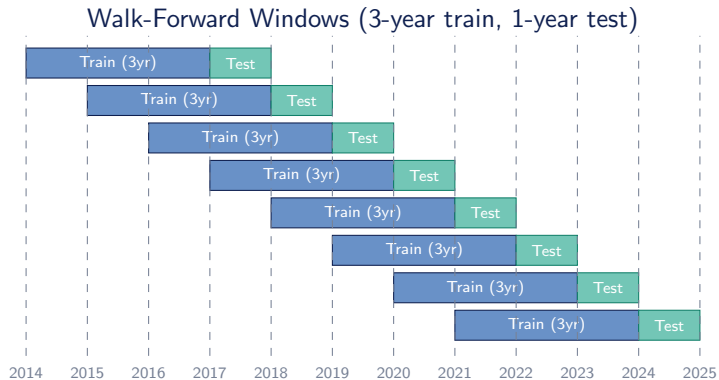
- Train a classifier to predict whether an asset will outperform the cross-sectional median at the next trading step(next day for daily rebalancing)
- Go long on the top k assets in terms of probability of overperforming and short on the bottom k .
- We used: A Random Forest, Gradient Boosted Tree and a 2 layer MLP for this approach

But these models are not designed for time-series data and do not account for temporal dependencies, how is this dealt with?

ML-Based Prediction: Feature Engineering

- Passing returns over an entire training year would require 252 input features, which is too high-dimensional for our sample size.
- Instead we use daily returns for the past 20 trading days, which gives us 20 features per stock. To account for long range dependencies, we also pass returns over 20 day intervals for the rest of the year, giving us 11 additional features. This results in a total of 31 features per stock(20 daily + 11 interval returns).
- **Failure of ARIMA and MomentFM:** We attempted to use time series models(ARIMA as a baseline) for this but they greatly underperformed.

Walk-Forward Evaluation



Baseline Strategies

Strategy	Description
40-60	40% cash at 6.5% p.a. + 60% tracking ^NSEI
EqualWeight	Daily rebalance to equal weight across all stocks
EW Buy&Hold	Equal weight on day 1, no rebalancing
Greedy	100% in the best-performing stock from the training window
Market	100% in ^NSEI (NIFTY 50)

Shared assumptions: Starting capital \$100 · Risk-free 6.5% p.a. (Indian T-bill) · Same walk-forward windows

Thank You

Questions & Discussion